Experiment 30: CART for Car Price Prediction

Aim: To predict used car prices using a Decision Tree Regressor and display the decision path.

Algorithm:

1. Import Decision Tree Regressor.
2. Create car mileage and price data.
3. Train the regression model.
4. Get new car details from the user.
5. Predict the car price.
6. Display the decision path.

Code:

```
from sklearn.tree import DecisionTreeRegressor


X = [[10000], [20000], [30000], [40000], [50000]]
y = [900000, 800000, 700000, 600000, 500000]


model = DecisionTreeRegressor(random_state=42)
model.fit(X, y)


mileage = int(input("Enter car mileage: "))


price = model.predict([[mileage]])
print("Predicted Car Price:", price[0])


path = model.decision_path([[mileage]])
print("Decision Path:", path.toarray())
```

Input:

```
Enter car mileage: 30000
```

Output:

```
Predicted Car Price: 700000
Decision Path: [[1 1 1 0 0 0 0 0 0]]
```



Result: The car price was predicted using CART, and its decision path was displayed.